INCH-POUND

MIL-R-49315(CR)
13 April 1990

MILITARY SPECIFICATION

RADIOSONDE SET ML-659(V), FAMILY OF

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 SCOPE. This specification covers the family of Radiosonde Set ML-659(V) meteorological instruments. The family consists of the following Radiosonde Sets:

ML-659(V)1	NAVAID Mode, LORAN Receiver
ML-659(V)2	NAVAID Mode, VLF/OMEGA Receiver
ML-659(V)3	RDF Mode

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-13231	- Marking of Electronic Items
MIL-F-14072	- Finishes for Ground Signal Equipment
MIL-R-46085	- Rhodium Plating, Electrodeposited
MIL-T-49447	- Temperature Sensor, Resistance

Benenficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey, 07703-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-H-49448 - Humidity Sensor, Resistance

STANDARD

MILITARY

MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-810 - Environmental Test Methods
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities

(Unless otherwise indicated, copied of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) the contractor shall furnish first article units for first article inspection and approval (see 4.3).

3.2 Construction. The Radiosonde Sets shall be constructed in accordance with the requirements of this specification. The common parts to all Radiosonde Sets shall consist of a case, electronic commutator, modulator, battery and meteorological sensors. In addition, each of the Radiosonde Sets shall include:

a. ML-659(V)1, NAVAID Mode, shall consist of a transmitter (400-406 Hz) with the corresponding antenna and LORAN receiver with the corresponding antenna.

b. ML-659(V)2, NAVAID Mode, shall consist of a transmitter (400-406 MHz) with the corresponding antenna, and VLF/OMEGA receiver with the corresponding antenna.

c. ML-659(V)3, RDF Mode, shall consist of a transmitter (1660-1700 MHz) with the corresponding antenna.

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The meteorological sensors shall be defined as follows:

a. Temperature sensor
b. Humidity sensor
c. Pressure sensor or baroswitch assembly

3.3 Parts, materials and processes. Parts, materials, processes and microcircuits shall be standard commercial of sufficient quality to meet requirements specified herein.

3.4 Recycled, virgin and reclaimed materials. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

3.5 Finish, protective. All surfaces of the Radiosonde Set shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware type items, such as handles, hinges, screws, etc., and necessary touch-up after mounting. This requirement shall not apply to the humidity or temperature sensor, nor to the baroswitch parts or plastic parts.

3.6 Marking.

3.6.1 General. Marking of Radiosonde Set shall be in accordance with MIL-M-13231, table II, group 3. (see 4.4)

3.6.2 Visibility. Whenever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipments.

3.6.3 Serial numbers. Each unit of Radiosonde Set shall be serial-numbered. (see 4.4)

3.6.4 Radiosonde identification notice. Each radiosonde shall have affixed to it an identification notice having the following text:

"This instruments is a radiosonde. It was used by the US Armed Forces to obtain high altitude weather data. It is not dangerous and is of no further use to the government. Do not return, dispose of it in any convenient manner."

3.7 System Safety Engineering.

3.7.1 Personnel Hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirements 1 and 8 shall be made

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applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection. (see 4.9)

3.7.2 Edge Rounding Exposed. Exposed edges and corners shall be rounded to a radius to prevent injury to personnel.

3.7.3 Radioactive Materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arresters and lenses).

3.7.4 Equipment Safety. Means shall be provided to minimize equipment damage, degration of efficiency or mission failure due to the following conditions:

a. Operator induced errors.

b. Improper cabling.

c. Power failure or electrical overstress components.

d. Secondary failures.

e. Installation, storage, operation, handling, maintenance, and transportation.

3.8 Human engineering. Design configuration as affecting operation and handling shall be maintained through compliance of configuration criteria to requirements of MIL-STD-1472.

3.9 Size and weight. The size of the Radiosonde Set shall be no greater than 17 in (43.2 cm) long by 5 in (12.7 cm) wide by 7 in (17.8cm) high. The Radiosonde Set weight, less battery, shall not exceed 24.7 oz (700g) and shall not exceed 35.3 oz (1 Kg) with battery. The battery weight shall not exceed 300 grams when dry.

3.10 Performance requirements. The Radiosonde Sets ML-659(V) shall measure and telemeter temperature, relative humidity and pressure data to the ground station for the Radio Direction Finding and NAVAID subsystems and receive and transmit NAVAID data for the NAVAID subsystem when required. The specified performance requirements for each part of Radiosonde Set shall be defined in the following paragraphs.

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3.10.1 Transmitter. The transmitter shall consist of 1 of 2 types; one operating at 403 MHz and one operating at 1680 MHz. The transmitter shall be a solid state device. The transmitter shall be FM. Themeteorological frequency deviation shall be 150 ± 30 KHz (unsymetrical) and NAVAID frequency deviation shall be 200 ± 40 KHz peak-to-peak.

3.10.1.1 403 MHz output.

3.10.1.1.1 Frequency. The transmitter shall be continuously tunable over the frequency range of 400 to 406 MHz and shall be set to 403 ± 0.5 MHz upon delivery. The transmitter shall be capable of being set to any frequency within the tuning range to an accuracy of ± 0.025 MHz.

3.10.1.1.2 Frequency stability. The transmitter shall be such that at any set frequency between 400 to 406 MHz and after a 10 minute stabilization period at ambient temperature and normal operating voltage, the frequency drift shall remain within a 1.0 MHz frequency band when subject to any service condition specified in 3.11 and any voltage between the minimum and maximum operating voltage of the battery.

3.10.1.1.3 Transmitter radiated power output. The performance of the transmitter at any frequency range from 400-406 MHz shall be such that after a stabilization period of 10 minutes at ambient temperature and normal operating voltage, the power output from the transmitter shall not be less than 180 milliwatts when subjected to any service operating condition specified in 3.11 and any voltage between the minimum and maximum operating voltage of the battery.

3.10.1.1.4 Antenna. The polarization shall be vertical and matched to the transmitter output with a VSWR not exceeding 1.5 to 1 over the frequency range of 400-406 MHz.

3.10.1.1.5 Superregeneration. The transmitter shall not superregenerate.

3.10.1.2 1680 MHz output.

3.10.1.2.1 Frequency. The transmitter shall be continuously tunable over the frequency range of 1660 to 1700 MHz and shall be set to 1680 ± 4 MHz upon delivery. The transmitter shall be capable of being set to any frequency within the tuning range to and accuracy of ± 0.025 MHz.

3.10.1.2.2 Frequency stability. The transmitter shall be such that any set frequency between 1660 to 1700 MHz and after a 10 minute

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stabilization period at ambient temperature and normal operating voltage, the frequency drift shall remain within a 2.0 MHz frequency band when subject to any operating service condition specified in 3.11 and any voltage between the minimum and maximum operating voltage of the battery.

3.10.1.2.3 Transmitter radiated power output. The performance of the transmitter at any frequency from 1660 to 1700 MHz shall be such that after a stabilization period of 10 minutes, at ambient temperature and normal operating voltage, the power output from the antenna shall not be less than 200 milliwatts when subject to any operating service condition specified in 3.11 and any voltage between the minimum and maximum operating voltage of the battery.

3.10.1.2.4 Antenna. The polarization shall be circular and matched to the transmitter output with a VSWR not exceeding 1.5 to 1 over the frequency range 1660-1700 MHz.

3.10.1.2.5 Superregeneration. The transmitter shall not superregenerate.

3.10.2. Modulator. The modulator shall consist of a meteorological data oscillator (MDO), a baroswitch assembly, and an electronic switch (electronic commutator).

3.10.2.1 Meteorological data oscillator. The meteorological data oscillator shall provide an output for frequency modulating the transmitter at a frequency which is a function of the meteorological sensor resistance and a pulse width is a function of the parameter being measured.

3.10.2.2. Pulse width and rise and fall times. The meteorological data oscillator shall have the following pulse widths as a function of the parameter being transmitted:

Parameter	Pulse width (microsecond)
Reference	425 ± 30
Humidity	615 ± 40
Temperature	900 ± 45
Pressure	1285 ± 50

The pulse rise and fall times shall be 140 to 175 microseconds between the 10 and 90 percent rise time points.

3.10.2.3 Modulation frequencies. The meteorological data oscillator with sensors switched into the input circuit shall operate over the frequency range of 12.5 to 450 Hz and at 475 Hz for the reference

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resistor at ambient temperature and nominal voltage (23.0 ± 0.5 volts).

3.10.2.4 Frequency stability The frequency stability of the meteorological data oscillator shall be such that maximum deviation in reference shall not be exceeded. The resistance to frequency requirement of 3.10.2.7 shall not be exceeded.

3.10.2.5 Meteorological data oscillator sensor current. The meteorological data oscillator sensor current over the frequency range of 12.5 to 450 Hz shall not exceed 75 microamps dc.

3.10.2.6 Leakage resistance (modulator). The leakage resistance of the modulator at high humidity shall not cause the open circuit frequency to exceed 1.0 Hz.

3.10.2.7 Meteorological data oscillator transfer characteristic. The meteorological data oscillator resistance vs frequency transfer characteristic shall be as follows after normalization of reference frequency:

Meteorological data oscillator Resistance (ohm)	Frequency Transfer (Hz)
0	475.0
7515	410.5
11240	384.6
17731	346.5
29768	292.7
53709	223.6
153577	112.6
352507	56.6

The transfer characteristic accuracy shall be within ± 0.5°C. The transfer characteristic accuracy shall be verified using the following equations:

EQUATION 1 R'_S or $R_t = R_{int}\ (f(F_S)-1)$

R'_S = The total resistance in the humidity circuit (The humidity sensor is in parallel with a 1.2 megohm resistor.)

R_t = The temperature sensor resistance (at t°Celsius).

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F_r = The frequency with zero resistance in the circuit (reference). The nominal value is 475 Hz.
F_s = The frequency with the sensor in the circuit.

$$f(F_s) = \frac{1}{\sum_{K=0}^{4} A_k (F_s/F_r)^k}$$

$A_0 = 0.0$
$A_1 = 1.0026$
$A_2 = -.0091$
$A_3 = .0091$
$A_4 = -.0026$

Equation 2

$$Rint = \frac{Rm}{f(F_m)-1}$$

$$f(F_m) = \frac{1}{\sum_{k=0}^{4} A_k (F_m/F_r)^k}$$

A_k constants are the same for Equations 1 and 2.

R^m = the resistance in the circuit for mid-reference. (the nominal value is 47775 ohms).
F^m = frequency with R^m in the circuit. The nominal value is 237.5. Hz

Equation 3

$$t = \frac{1}{\sum_{k=0}^{3} A_k (\ln(R_t/R30))^k} - 273.15$$

t = The measured temperature (degree Celsius)
R_t = The temperature sensor resistance (at any t^oC.)
R_{30} = The temperature sensor lock-in resistance at $+30^oC$.

A_0 =	3.2987 E-03
A_1 =	4.7764 E-04
A_2 =	3.0029 E-06
A_3 =	1.5108 E-06

3.10.2.8 Harmonic content. The harmonic content of the

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meteorological data oscillator including the electronic clock, shall be limited to a level that does not cause degradation of the processed NAVAID data in the AN/TMQ-31 Meteorological Data System.

3.10.2.9 Meteorological data oscillator mid-scale correction. A mid-scale correction shall be accomplished whenever the baroswitch is on an insulated commutator segment. The value of the mid-scale resistor shall be 47,775 ± 0.1 percent ohms.

3.10.3 Radiosonde battery. The Radiosonde Set battery shall be of the magnesium cuprous chloride water activated type, Battery BA-852(), having a minimum and maximum voltage of 15.0 and 23.0 volts dc, respectively. The nominal voltage is 18.0 ± 0.5 volts dc.

3.10.3.1 Activation time. The time to activate the battery shall not exceed 10 minutes.

3.10.3.2 Minimum time to end voltage. The minimum time to end voltage shall not be less that 120 minutes. This shall include a 10 minute discharge time at surface temperature and pressure and a 110 minute discharge time under pressure-temperature flight similitude to an altitude of 105,000 feet (32,000 meters). The pressure-temperature rate of change shall follow that of the standard atmosphere for an increase in altitude at a rate of 1000 feet per minute (305 meters per minute).

3.10.4 Receiver. The receiver shall consist of an antenna and amplifier. The receivers shall be utilized to receive and amplify LORAN-C/D, and OMEGA/VLF/VLF NAVAID data. The receiver shall be a solid-state device.

3.10.4.1 LORAN C/D.

3.10.4.1.1 Antenna. A 100 KHz antenna shall be provided and matched to the amplifier input with a VSWR not exceeding 1.5 to 1.

3.10.4.1.2 Amplifier gain/bandwidth. The amplifier gain/bandwidth shall be as shown in Fig 1.

3.10.4.2. OMEGA/VLF.

3.10.4.2.1 Antenna. An antenna shall be provided and matched to the amplifier input with VSWR not exceeding 1.5 to 1 at 12 KHz.

3.10.4.2.2 Amplifier gain/bandwidth. The amplifier gain/bandwidth shall be as shown in Fig 2.

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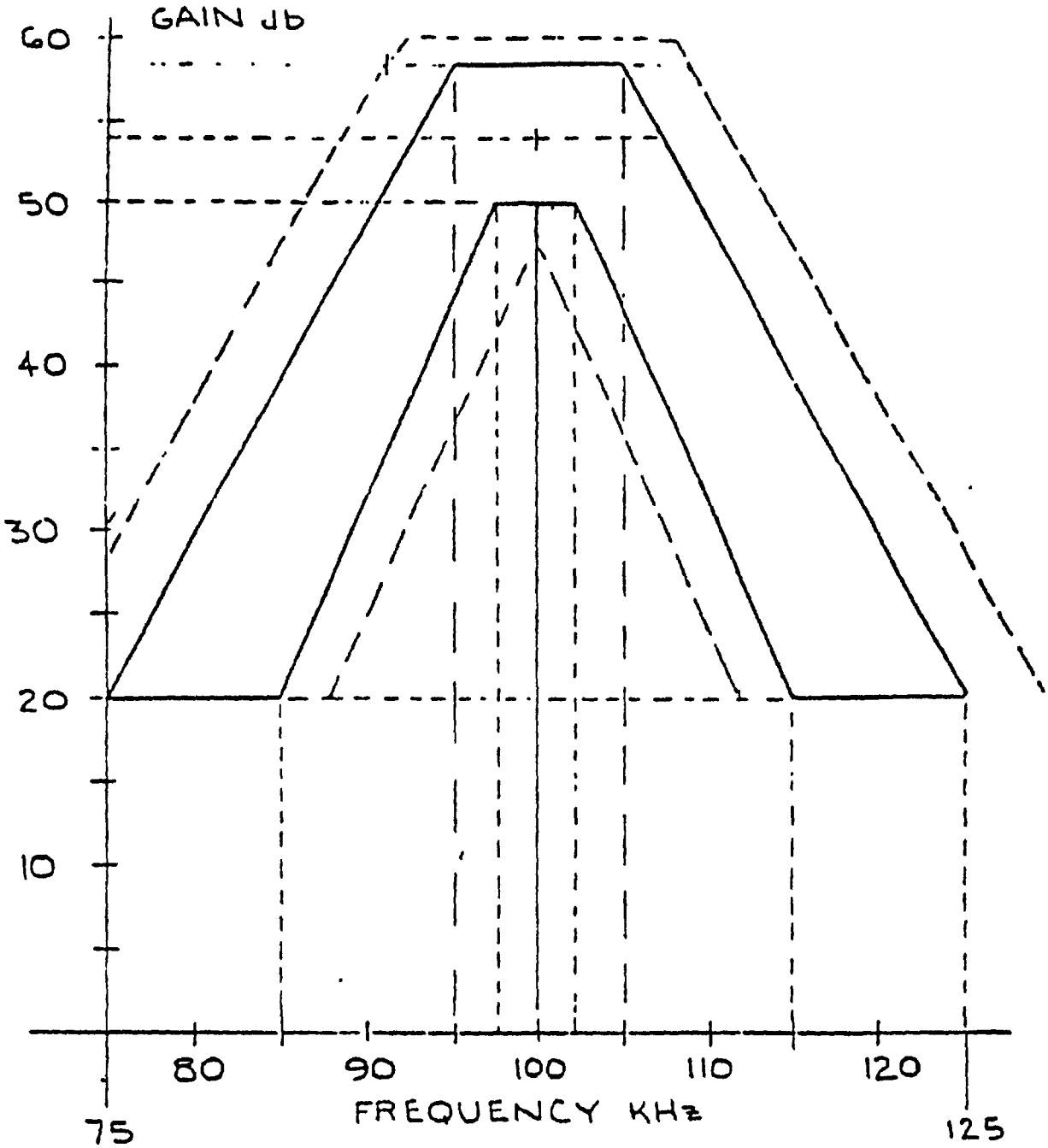


Fig. 1. LORAN acceptance profile MDS receiver.

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Note: Ambient conditions.

1. Gain: 54 ± 4 dB

2. Profile:

58 dB	Lowside	95 KHz
	Highside	105 KHz
50 dB	Lowside	97.5 KHz
	Highside	102.5 KHz
20 dB	Lowside	75 KHz minimum
		85 KHz maximum
20 dB	Highside	115 KHz minimum
		125 KHz maximum

Adverse environmental condition.

3. Gain: 54 ± 6 dB

4. Profile:

60 dB	Lowside	91 KHz
	Highside	109 KHz
48 dB	Lowside	100 KHz
	Highside	100 KHz
20 dB	Lowside	71 KHz minimum
		87.5 KHz maximum
20 dB	Highside	112.5 KHz mimimum
		129 KHz maximum

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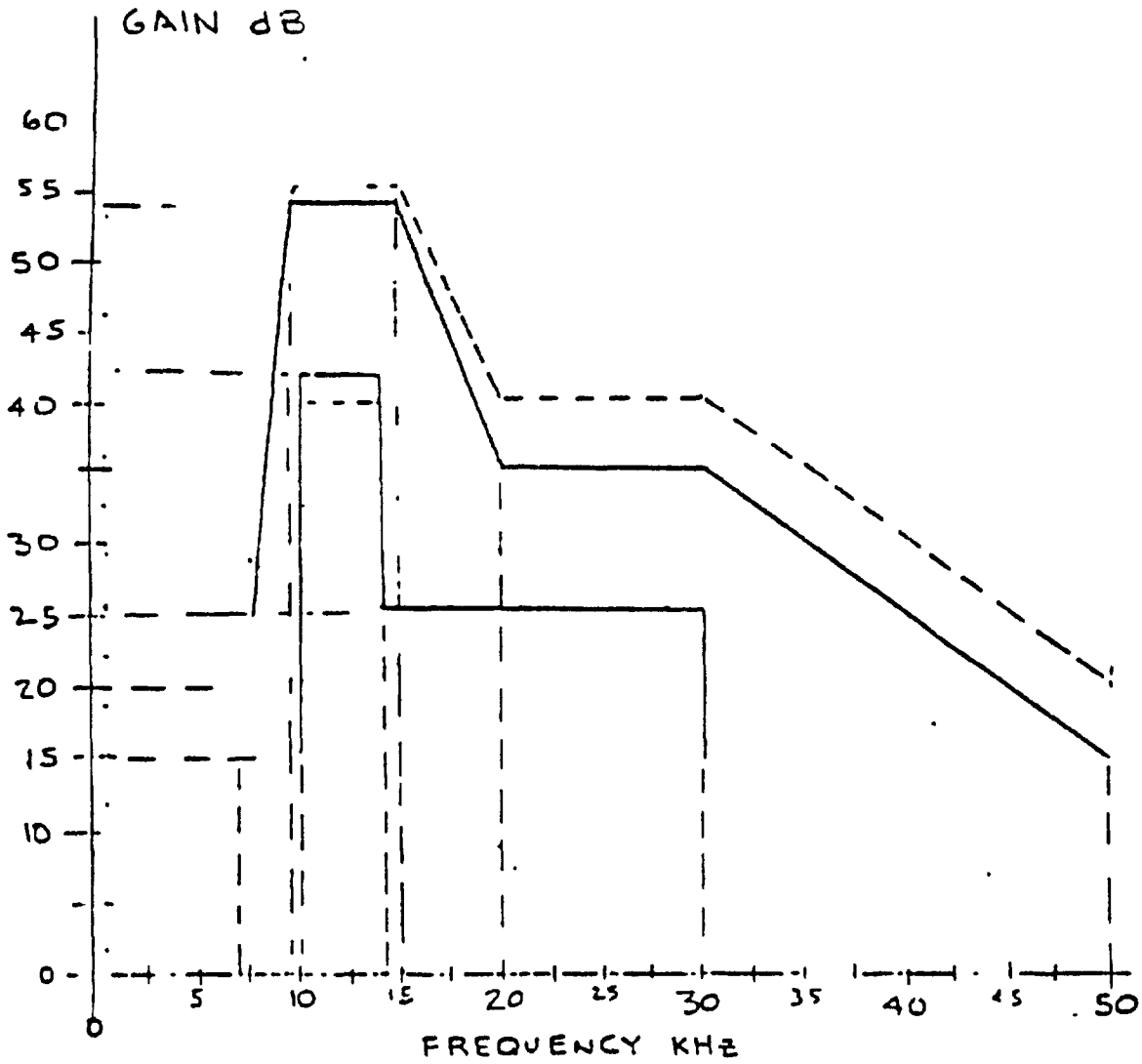


Fig. 2. VLF/OMEGA acceptance profile - MDS

NOTES: Ambient profile turning points

1. Maximum gain:

Gain	KHz
25	7
54	9 to

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54	15
35	20 to
35	30
25	40

2. Mimimum gain:

Gain	KHz
42	10 to
42	14
25	14 to
25	30

Adverse environment profile turning points.

3. Minimum gain

Gain	KHz
28	7
55.5	9 to 15
40	20 to 30
27	40

4. Minimum gain

40	10 to 14
25	14 to 30

3.10.5 Baroswitch assembly. The baroswitch shall operate over the pressure range between 1060 millibars and 5 millibars (mb).

3.10.5.1 Aneroid capsule. The aneroid pressure capsule shall be fabricated from "Ni Span C" alloy, aged under ambient conditions for 30 calendar days prior to calibration to insure that the capsule has attained a stabilized condition. The date of fabrication shall be legibly and indelibly marked on the capsule.

3.10.5.2 Linkage mechanism. The pressure capsule diaphragm, push pin and contact arm shall be properly aligned so as to miminize friction. The merchanism shall be effectively spring-loaded so as to reduce backlash to an absolute minimum. The mechanism shall operate properly at any temperature at which the radiosonde will be used. The entire mechanism shall be constructed so as to preclude any disengagement of any of the components.

3.10.5.3 Baroswitch deflection. The baroswitch shall have essentially a logarithmic relationship of deflection versus pressure such that there will be less than 20 millibars (mb) per contact at 1000 mb and

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less than 4 mb per contact at 50 mb. The push pin shall not disengage from the pressure capsule when the arm is moved to the 179th contact at sea level pressure.

3.10.5.4 Setting error. The setting error is defined as the difference between the true pressure and the pressure indicated by the baroswitch, as obtained from the pressure calibration chart or tape, at the beginning of the sixth contact, when the pressure is decreasing. This difference is considered a "positive" error when the indicated pressure is higher than the true pressure and "negative" when less than the true pressure. This setting error shall not exceed ± 4 millibars

3.10.5.5 Calibration accuracy. The baroswitch, at 27°C ± 7°C, and after algebraic corrections for the setting error, shall provide an accuracy of ± 4 millibars at pressure in excess of 150 mb, and an accuracy of ± 2 mb at pressure of 150 mb or less. NOTE: Corrected chart pressure equals the indicated chart pressure minus the setting error times A/B.

Where: A = Pressure interval at any given contact
and B = Pressure interval of 6th contact

The pressure interval is defined as the width of the contact in millibars.

3.10.5.6 Low temperature effect. When the baroswitch is subjected to a temperature which is 80 ± 30°C below the room temperature run of 3.10.5.5 its calibration run pressure shall agree with the room temperature calibration to within ± 4 mb at pressures in excess of 150 mb, and to within ± 2 mb at pressure of 150 mb or less.

3.10.5.6.1 Baroswitch offsetting. When the control of the aneroid capsule manufactured is such that cold compensation is under statistical control, the manufacturer may provide a bias to the original calibration chart for improved accuracy at high altitudes. This shall be done by programming the computer to compensate for the average cold error. When this compensation is added to the original chart; paragraphs 3.10.5.5 and 3.10.5.6 shall be changed as follows:

The acceptance shall be based on the range from 1050 mb to 250 mb when the rerun is at 27°C + 7°C and over the range 750 mb to 5 mb when run is at 80°C below ambient. There are no changes in the acceptance limits.

3.10.5.7 Friction. The deviation in observed pressure between the make point of a conducting segment under decreasing pressure conditions

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and the break point of the same conducting segment when the pressure is increasing shall not exceed the limits specified in (a) and (b) below. The rate of change of pressure shall be equal to or grater than the normal sonde ascent rate (1000 feet/minute) (305 meters/minute). The unit shall be subjected to the full operating range of pressure change, first decreasing, then increasing the pressure at the rate specified above.

	Range	Max Deviation
a.	For pressure in excess of 150 mb	10 mb
b.	For pressure less than 150 mb	6 mb

3.10.6 Detent operation. The adjustment of the commutator in relation to the contact point shall have a range of not less that 10 contacts at sea level pressure. The commutator adjustment shall be accomplished by means of a detent contact screw such that each "click" of the adjustment knob shall move the commutator, relative to the contact point, a distance equivalent to a 0.5 ± 0.1 mb pressure change, in pressure range of 1000-950 mb. Repeatability of the detent adjustment shall be within ± 1 "click"

3.10.7 Bearing tip. The contact point which bears on the commutator shall be made of heat treated, hardened, beryllium-copper, gold plated with a final plating of rhodium Type 1, Class II, per MIL-R-46085. The bearing tip of the point shall be polished in order to meet the operational requirements.

3.10.8 Contact pressure. The contact point shall remain in contact with the commutator when lifting force of 1.0 gram is exerted at the tip of the contact arm immediately adjacent to the contact. Contact between the contact point and the commutator shall be broken when a lifting force of 2.5 grams is applied to the tip to the contact arm immediately adjacent to the contact point. This shall apply to any point along the commutator.

3.10.9 Contact resistance. The baroswitch contact resistance shall not exceed 10 ohms.

3.10.10 Contact arm support. The contact arm support of the baroswitch shall hold the bearing tip off the commutator from contact 0 to contact 179 during shipment. The operation of the support shall be positive in both position so that the arm will not be raised from the

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commutator during operation.

3.10.11 Commutator. The pressure sensor commutator configuration shall consist of 179 contacts comprised of 12 groups of 7 segments and 1 group of 5 segments with pressure identifiers interspersed with intermediated contacts The surface of the commutator shall be polished smooth.
The conducting segment shall be made of electro-deposited silver and, when necessary, an additional surface of electro-deposited gold. The insulating segments shall be fabricated from a suitable molding material (thermoplastic or thermosetting plastic) capable of meeting all specification requirements when exposed to all service conditions.

3.10.11.1 Commutator sequence.

The sequencing of the commutator shall be as follows:

Contact No.	Output
1	R0
2	R6
3	1/
4	R5
5	1/
6	R3
7	1/
8	R1
9	1/
10	R0
11	1/
12	R2
13	1/
14	R4
15	1/
16 Repeat of 1st group	R6
17 " " " " "	1/
18 " " " " "	R5

1/ Intermediate Contact

Resistance values for the various R values shall be as follows:

	Resistance Ohms		
R Value	Low(+950)	Nominal	High
R0	0	0	50

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R Value	Resistance Ohms Low(+950)	Nominal	High
R1	500	1387	1937
R2	1996	2687	3537
R3	3629	4696	6009
R4	6151	7800	9828
R5	10.047K	12.594K	15.727K
R6	16.065K	20.000K	40.000 1/

1/ Value shall be limited to produce a resistance less than the midscale resistance (47,775 ohms).

3.10.11.2 Adherence of conducting segments. The adherence of the conducting segments to the plastic material shall be positive.

3.10.11.3 Leakage resistance (commutator). The commutator leakage resistance, after cleaning and polishing the commutator, between metallic ground and all conducting segments of the commutator connected together, shall not be less than 10 megohms.

3.10.11.4 Segment width and thickness. The spacing between the leading edge of successive conducting segments of the assembled commutator shall be the same and the width of insulated segments shall be sufficient to preclude shorting segments by the contact point.

3.10.11.5 Flight termination. Means shall be provided for terminating the radiosonde flight by opening the B+ circuit to the transmitter. The termination altitude shall be selectable at up to ten pressures on the commutator. The lowest selectable altitude shall be not greater than 6 kilometers. The highest selectable altitude shall be not less than 20 kilometers.

3.10.12 Baroswitch calibration chart. A baroswitch calibration chart showing atmospheric pressure in millibars versus contact numbers, in the form of a digital chart or table shall be provided. The chart shall be made of the automatic system to provide uniformity. The initial contact shall represent a pressure not more than 1088 nor less than 1060 millibars and the final contact not greater than 5 millibars, distributed over at least 155 contacts. Each calibration chart shall be printed or otherwise indelibly marked with the serial number of the baroswitch frame, and the click detent value of the detent adjusting screw. The calibration chart of the instrument shall be included together with the packed instrument. Any irregularity of differential pressure between successive contacts shall not exceed 2 millibars nor exceed 1 mb for pressures less than 350 millibars. The pressure altitude in kilometers of the 10 termination altitudes shall be

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included on the calibration chart. The chart shall show the value of the R6 resistor for contact groups 1 and 2.

3.10.13 Baroswitch punched paper tape calibration. A baroswitch punched paper tape calibration shall be provided with the same information required as the same information required as that for the baroswitch calibration in 3.10.12. The baroswitch frame serial number shall be marked at the leading edge of the tape. Capability for manual entry of baroswitch calibration shall be provided. The tape format shall be in accordance with Figures 3 and 4. The punched paper tape shall be packaged together with the corresponding calibration chart.

3.10.14 Instrument case. The instrument case shall be opaque and white and provide a rainproof space for all components and a battery pack. It shall also provide mountings for the meteorological sensing elements and a supporting ring for attachment to a balloon cord. Insulation shall be provided between the sensor housing and the battery compartment. The case shall no warp or become destorted.

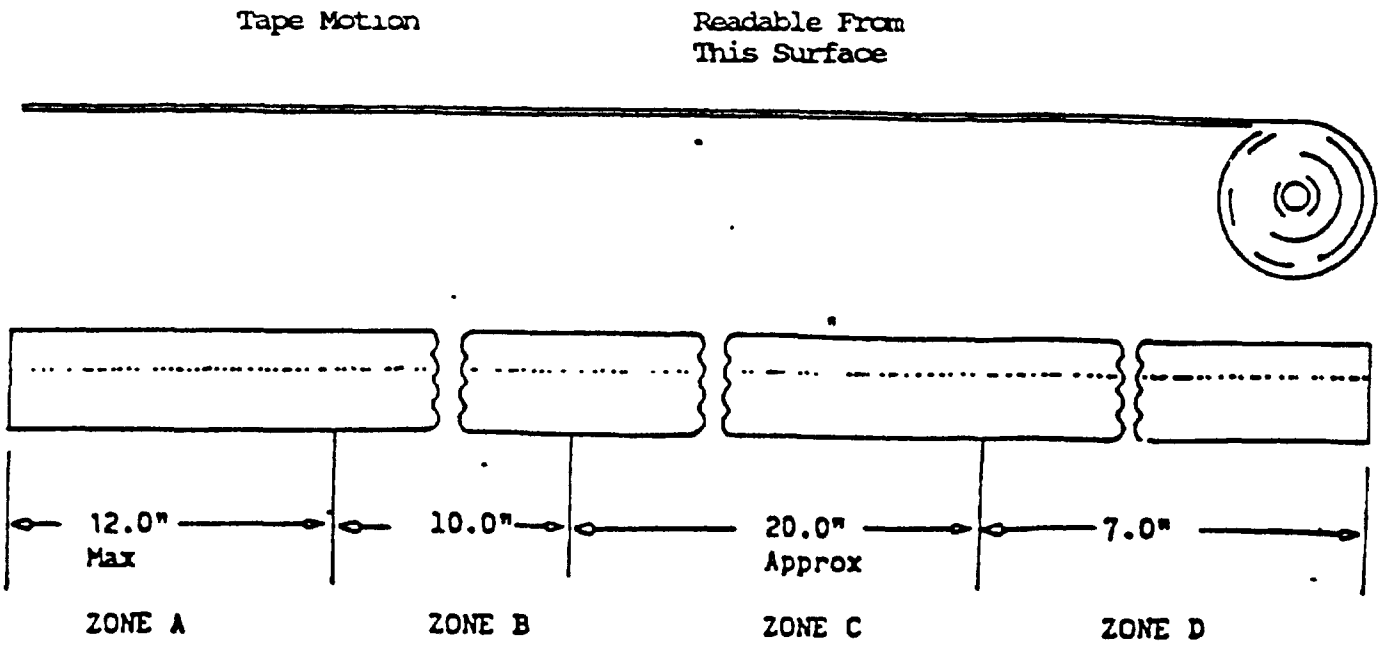


Figure 3. Punched paper tape format.

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NOTES:

1. ZONE A - Shall be an eye readable section consisting of 3 filler characters followed by the baroswitch serial numbers, in the format of XXX-XXXX, followed be 3 filler characters.

2. ZONE B - Shall consist of 10 inches of drive holes only.

3. ZONE C - Shall consist of three subzones as follows:

C1. Machine readable baroswitch serial number in ASCII with even parity, in the format XXX-XXXX where the fourth frame will always be a "-"(ASCII code 55 octal). This subfield will end with "ETX" (ASCII code 30 octal) and will not be included in the checksum. (C3 below)

C2. Pressure values for all contacts expressed in 182 binary 8 bit fields without parity. Data shall be provided for 180 contacts with pressure value in 0.1 mb expressed as a difference from the pressure of the previous contact. The first contact uses 3 frames whose binary sum is the required data with 1088.0 mb assumed to be the previous pressure. If any null frames are used they should occur first.

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Data for each additional contact shall occupy one frame. If there are less that 180 active contacts, the remaining frames shall be filled with binary zero.

C3. A binary checksum of all frames, expressed in two frames, most significant frame first.

4. ZONE D - Shall consist of 7 inches of drive holes only.

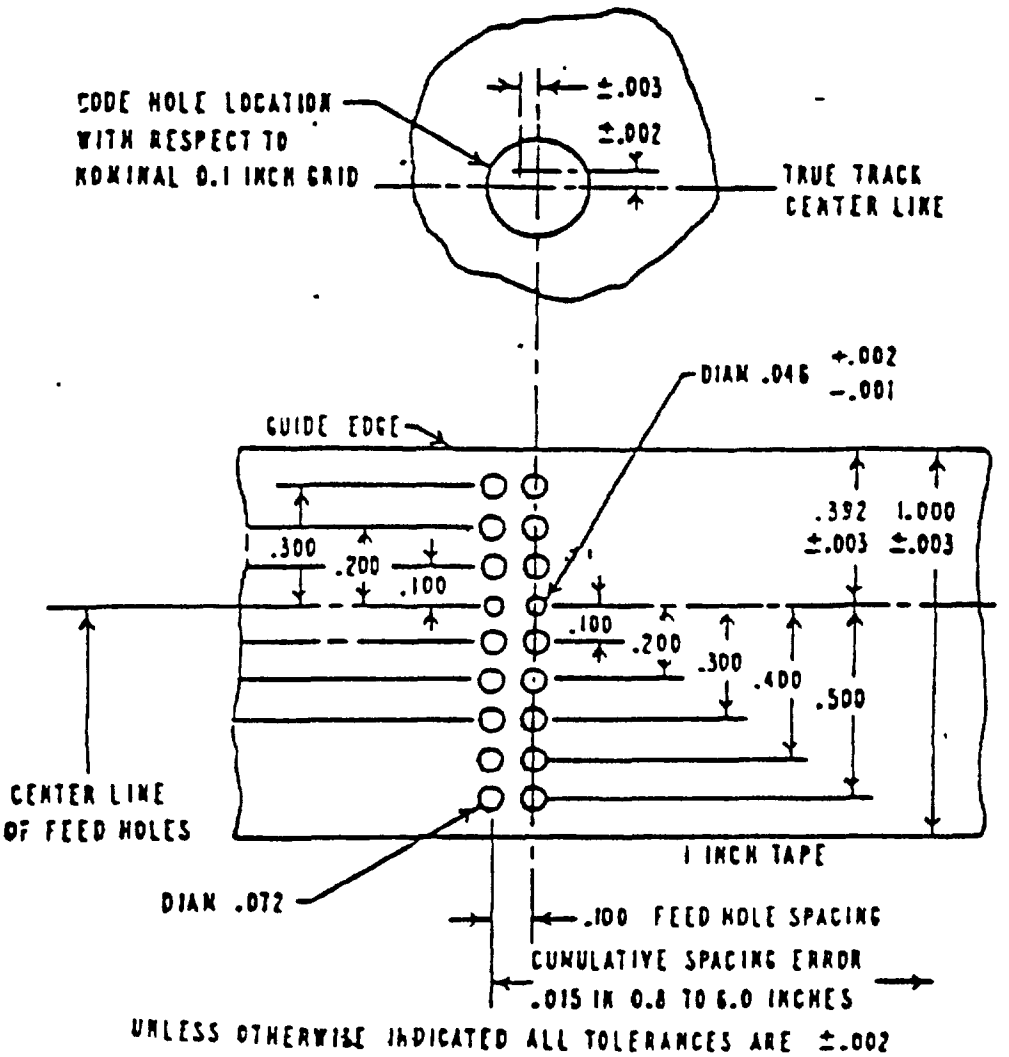


FIGURE 4. PUNCHING STANDARD FOR PAPER TAPES

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3.10.14.1 Reflectivity. The material used for the upper assembly of the case, where the sensors are mounted, shall have a reflectivity to solar radiation of 85 percent in the visible light region, when compared to a freshly scraped magnesium carbonate block, or equal.

3.10.14.2 Surface treatment. The surface of the case between the humidity element clips and the temperature element clips and the support arms shall be so treated as to resist wetting by rain water, or manufactured of a material possessing nonwetting properties.

3.10.14.3 Humidity sensor housing. A reflective surface coating shall be applied over those exterior surfaces of the humidity element housing which become exposed to solar radiation. The reflective surface coating shall cover at least 90 percent of the potentially exposed area and shall have an absorbance factor not grater than 0.25 in the energy band from 0.30 to 4.0 microns. The interior surfaces shall be coated with a black polystyrene.

3.10.14.4 Strength of case, ring and harness. The supporting ring, harness, and case attachment shall withstand a 25 pound pull.

3.10.14.5 Humidity element mounting. The humidity element shall be mounted so that it is sheltered from precipitation, but exposed to an airflow of at least 85 percent of that if mounted without any obstructions. The element mounting shall be constructed from spring tempered phosphor bronze Type A wire.

3.10.14.6 Temperature element mounting. The temperature element shall be mounted so the it is exposed to the atmosphere and, yet, sufficiently removed from the case to reduce any temperature error due to heating from the case.

3.10.14.7 Temperature element clips. The clips for the temperature element shall be capable of securely holding a temperature element lead when subjected to a pull of at least 175 grams. The clips shall be insulated from the support arms with a "non-wetting" material at least 1 inch long, having a leakage resistance of a least 100 megohms.

1.10.15. Meteorological sensors.

1.10.15.1 Temperature sensor. The temperature sensor shall be in accordance with MIL-T-49447(CR) and shall be supplied with sufficient data to eliminate the need for a baseline check set.

3.10.15.2 Humidity sensor. The humidity sensor shall be in accordance with MIL-H-49448 (CR). A 1.2 megohm ± 2 precent resistor

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shall be connected in parallel with the humidity element. Sufficient data shall be supplied with each element to eliminate the need for a baseline check set.

3.10.16 Electronic commutator. The electronic commutator shall sequentially switch into the modulator circuit, the temperature and humidity sensors, pressure resistors, and reference resistor.

3.10.16.1 Commutator sequence. The commutator sequence shall be P, T, P, H, P, T, P, R.

P = Pressure; T = Temperature; H = Humidity; R = Reference

3.10.16.2 Commutator ON time. The ON time shall be 1 ± 0.1 seconds.

3.10.16.3 Commutator switch resistance variation. The maximum variation in "ON" resistance for the commutator shall not exceed 40 ohms.

3.10.17 Test connector with leads. A test connector with test leads attached shall be provided for ground checking of specific internal parameters. The leads shall be 4 ± 1 inches in length, shall be of AWG number 22 or number 24 standard wire and shall be stripped and tinned 0.75 ± 0.25 inches from their ends.

3.10.17.1 R6 test leads. Test leads, blue and white, when shorted together shall cause the Radiosonde to transmit the R6 value for the first group when the baroswitch contact arm is in the storage position.

3.10.17.2 Electronic commutator stop test leads. The electronic commutator stop test leads, brown and grey, when shorted together shall cause the Radiosonde to transmit pressure values only.

3.11 Service conditions. The Radiosonde Sets shall meet the following service conditions. (see 4.8)

3.11.1 Temperature-altitude. The Radiosonde Sets shall be operable within specified performance limits at all altitudes from sea level to 105,000 feet (32,000m). The corresponding temperature and solar radiation values are (see 4.8.1):

a. The Radiosonde Sets shall operate at ambient air temperatures from +125°F (+51.7°C) and 360 BTU per square foot per hour to -130°F (-90°C) and 442 BTU per square foot per hour.

b. The Radiosonde Sets in transportation or storage, nonoperating mode shall withstand air temperature in the range -70°F (-56.7°C) to +168°F (+75.6°C).

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3.11.2 Humidity. The Radiosonde Sets shall be operable without degradation in specified performance and shall sustain no physical damage during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas. (see 4.8.2)

3.11.3 Rain. The Radiosonde Sets shall be operable without degradation in specified performance and shall sustain no physical damage during periods of heavy precipitation with intermittent wind. (see 4.8.3)

3.11.4 Vibration and shock. The Radiosonde Sets shall withstand vibration and shock induced during field transport by military vehicle and by rough handling. The equipment shall withstand vibrations encountered during servicing. The Radiosonde Sets shall operate without degradation in specified performance and shall not suffer any mechanical damage after being subjected to the vibration test specified in 4.8.4.

3.11.5 Bench handling. The Radiosonde Sets shall operate without degradation in specified performance and shall suffer no mechanical damage after being subjected to the bench handling test specified in 4.8.5.

3.11.6 Fungus. After the fungus test of 4.8.6, the Radiosonde Sets shall be visually examined. The Radiosonde Sets shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10 percent or less of the total area and no more than 6 unrelated minute colonies with mycelial development in areas only in other than critical circuit portions; such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. In addition, the Radiosonde Sets shall operate without degradation in specified performance after exposure to the test conditions of 4.8.6.

3.11.7 Orientation. The Radiosonde Sets shall be operational in any orientation up to 90 degrees from vertical position. (see 4.8.7)

3.12 Reliability.

3.12.1 Reliability of radiosonde. The Radiosonde Sets, with battery shall meet a reliability requirement of 0.9 at 90 percent confidence level for a completion of an 11 Km sounding. A binomial failure distribution shall be assumed (see 4.5.5) with a minimum acceptable value of 0.90 and specified value of 0.96.

3.12.2 Reliability of battery. The battery shall meet a reliability requirement of 0.95 at 90 percent confidence level for completion of a

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120 minutes mission including a 10 minute activation time under the condition described in 3.10.3.2.

3.13 Workmanship. The Radiosonde Sets shall be manufactured and assembled in accordance with Requirements 5 and 9 of MIL-STD-454. (see 4.9)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3); does not include packaging.

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspection of equipment before packaging (see 4.5).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish 135 first article units of the Radiosonde Sets for groups A, B, C and D testing.

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4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering items listed in 4.4 and the inspections specified for group A, group B, group C and group D. The inspections of table 1 shall be performed in the following order: 4.4, group A (all units), group B (all units) designated units within group C in accordance with 4.5.3 and group D. After completion of group C environmental tests, conforming units shall be reinspected and shall pass all group A inspection. Failure of any inspection shall be cause for rejection.

4.3.3 First article data. The first article test plan and test reports shall be as required in the contract.

TABLE 1. First article inspection.

Inspections	Reqt para	Test para	Order of tests			
			Unit 1	Unit 2	Unit 3	Units 4-135
1. Inspections covered by subsidiary documents 1/	3.3 3.5 3.6	4.4	Inspection to be performed on all units			
2. Group A inspection 1/	See table II		Tests to be performed on all units			
3. Group B inspection 1/	See table III		Tests to be performed on all units			
4. Group C inspection						
Temperature-altitude	3.11.1	4.8.1	4			
Humidity	3.11.2	4.8.2	5			
Vibration and shock	3.11.4	4.8.4		4		
Bench handling	3.11.4	4.8.5		5		
Rain	3.11.3	4.8.3		6		
Orientation	3.11.7	4.8.7	6			
Fungus 2/	3.11.5	4.8.6			4	
Case reflectivity 2/	3.10.14.1	4.7.9.1	1	1	1	
Humidity sensor/housing 2/	3.10.14.4	4.7.9.2	2	2	2	
Humidity element mounting 2/	3.10.12.6	4.7.9.3	3	3	3	
5. Group D inspection						
Reliability	3.12	4.5.5				1

1/ The inspection 1 to 3, in the order shown, shall be performed on all

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first article units before subjecting these units to any other inspection requirements.

2/ Test required during first article test only.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable documents as part of the inspection required by this specification, and the inspection requirement specified in the contract.

Item	Where required
Parts, materials, and processes; general	3.3
Finish	3.5
Marking	3.6

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.4, 4.5.1 through 4.5.4 This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection Each unit on contract shall be inspected for conformance to the inspection in table II.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the government.

TABLE II Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Visual and mechanical	3.13	4.9
Baroswitch contact pressure	3.10.8	4.7.6.7
Commutator segment adherence	3.10.11.2	4.7.7.2
Commutator segment width and thickness	3.10.11.4	4.7.7.4
Flight termination	3.10.11.5	4.7.7.5

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TABLE II Group A inspection. (cont'd)

Inspection	Requirement paragraph	Test paragraph
Baroswitch paper tape	3.10.13	4.7.7.7
System safety	3.7	4.9
Performance		
Frequency (403 MHZ - Transmitter)	3.10.1.1.1	4.7.1.1.1
Frequency Spectrum (403 MHZ Transmitter)	3.10.1.1.6	4.7.1.1.6
Frequency (1680 MHZ - Transmitter	3.10.1.2.1	4.7.1.2.1
Frequency spectrum (1680 MHz transmitter	3.10.1.2.6	4.7.1.2.6
MDO modulation frequencies	3.10.2.3	4.7.2.2
MDO mid-scale correction	3.10.2.9	4.7.2.7
Baroswitch detent operation	3.10.6	4.7.6.6
Baroswitch contact resistance	3.10.9	4.7.6.8
Commutator sequence	3.10.11.1	4.7.7.1
Electronic commutator sequence	3.10.16.1	4.7.8.1
Electronic commutator switch resistance variation	3.10.16.3	4.7.8.3

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have been subjected to and have passed group A inspection. Each sample unit shall be inspected for conformance to the inspection in table III. The inspection of temperature sensor performance is

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a separate part of the group B inspection. If only the temperature sensor fails group B inspection, then only the temperature sensor shall be resubmitted for group B inspection.

4.5.2.1 Order of inspection within group B. Group B inspection shall be performed in an order satisfactory to the government.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test paragraph
Size and weight	3.9	4.6
Radiosonde performance		
Transmitter FM deviation	3.10.1	4.7.1
Frequency stability (403 MHz transmitter)	3.10.1.1.2	4.7.1.1.2
Radiated power output (403 MHz transmitter)	3.10.1.1.3	4.7.1.2.3
Superregeneration (403 MHz transmitter)	3.10.1.1.5	4.7.1.1.5
Antenna	3.10.1.1.4	4.7.1.4
Frequency stability (1680 MHZ Transmitter)	3.10.1.2.2	4.7.1.2.2
Radiated power output (1680 MHZ transmitter)	3.10.1.2.3	4.7.1.2.3
Superregeneration (1680 MHz Transmitter)	3.10.1.2.5	4.7.1.2.5
Antenna	3.10.1.2.4	4.7.1.2.4
Leakage resistance (modulator)	3.10.2.6.	4.7.2.1
MDO Pulse width and rise time	3.10.2.2	4.7.2.5
MDO Frequency stability	3.10.2.4	4.7.2.3

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TABLE III. Group B inspection (continued)

Inspection	Requirement paragraph	Test paragraph
MDO Transfer characteristic	3.10.2.7	4.7.2.4
MDO Sensor current	3.10.2.5	4.7.2.6
Amplifier gain/bandwidth (LORAN C/D)	3.10.4.1.2	4.7.4.1.2
Antenna	3.10.4.1.1	4.7.4.1.1
Amplifier gain/bandwidth (OMEGA/VLF)	3.10.4.2.2	4.7.4.2.2
Antenna	3.10.4.2.1	4.7.4.2.1
Baroswitch deflection	3.10.5.3	4.7.6.1
Setting error	3.10.5.4	4.7.6.2
Calibration accuracy	3.10.5.5	4.7.6.3
Low temperature effect	3.10.5.6	4.7.6.4
Friction	3.10.5.7	4.7.6.5
Leakage resistance (commutator)	3.10.11.3	4.7.7.3
Baroswitch calibration chart Electronic commutator	3.10.12	4.7.7.6
ON time	3.10.16.2	4.7.8.2
Strength of case, ring and harness	3.10.14.4	4.7.9.4
Holding capability of clips	3.10.14.7	4.7.10.1
Leakage resistance (support arms)	3.10.14.7	4.7.10.2

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TABLE III. Group B inspection (continued)

Inspection	Requirement paragraph	Test paragraph
Temperature Sensor performance		
Sensor reflectivity	3.10.15.1.1	4.7.5.1
Sensor humidity resistance	3.10.15.1.2	4.7.5.2
Sensor polarization	3.10.15.1.3	4.7.5.3
Thermal Shock	3.10.15.1.4	4.7.5.4
Sensor strength	3.10.15.1.5	4.7.5.5
Sensor resistance	3.10.15.1.6	4.7.5.6
Sensor resistance-temperatre	3.10.15.1.7	4.7.5.7

1/ All performance defect are major.

4.5.3 Group C inspection. Group C inspection shall be performed on inspection lots that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for group C inspection. Two Radiosonde Sets shall be selected at random, one sample for each subgroup I and II in table IV, from every 1000 units or fraction thereof produced. The first samples selected at the start of the contract shall be from the first quality conformance inspection lot.

4.5.3.2 Group C failures. Action required relative to group C failures shall be as specified in the contract.

4.5.3.3 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is required and the sample units are resubjected to and pass group A inspection only. Disposition of these samples shall be as states in the contract.

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TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Temperature-altitude	3.11.1	4.8.1
Humidity	3.11.2	4.8.2
Orientation	3.11.7	4.8.7
Subgroup 2		
Vibration and shock	3.11.4	4.8.4
Bench handling	3.11.5	4.8.5
Rain	3.11.3	4.8.3

4.5.3.4 Performance verification for group C sample units. During group C inspection performance measurements shall be in accordance with 4.8. At the completion of the group C inspection, all sample units shall be subjected to the following performance verification tests at ambient temperature:

Radiated power output (403 MHz)	4.7.1.1.3
Radiated power output (1680 MHz)	4.7..1.2.3
Transmitters (FM deviation)	4.7.1
Transfer characteristic	4.7.2.4
Amplifier gain/bandwidth (LORAN C/D)	4.7.4.1.2
Amplifier gain/bandwidth (OMEGA/VLF)	4.7.4.2.2
Calibration accuracy	4.7.6.3
Commutator ON time	4.7.8.2

4.5.4 Group D inspection. This inspection shall consist of the test specified in table V and shall be performed on units from lots which have been subjected to and met group A and group B inspections.

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TABLE V. Group D inspection.

Inspection	Requirement paragraph	Test paragraph
Reliability of radiosonde	3.12.1	4.5.5
Reliability of battery	3.12.2	4.5.5

4.5.5 Reliability testing.

4.5.5.1 First article and periodic group D.

4.5.5.1.1 First article reliability. To determine compliance with 3.12.1, 93 successful radiosonde flights to 11 Km altitude with no more that 6 relevant failures shall be performed. To determine compliance with 3.12.2, 45 batteries shall be tested in accordance with 4.7.3.1 and 4.7.3.2. This test shall be conducted in accordance with the government approved test plan and procedures as required in the contract.

4.5.5.1.2 Periodic group D inspection. Production units shall be consolidated into lots of 1,000 units for periodic group D inspection. A sample of 132 units shall be selected from each lot for test flights to an 11 km altitude. A sample of 45 batteries from every third lot shall be selected for periodic group D inspection of the battery.

4.5.5.2 Test conditions. The tests shall be conducted at a location specified in the contract. Balloons, launch equipment and hydrogen generator equipment shall be furnished by the government. Ground station receiving/monitoring equipment shall be provided by the contractor. Battery test shall be conducted under the environmental conditions specified in 4.7.3.2.

4.5.5.3 Performance measurements.

4.5.5.3.1 Performance measurements of radiosonde. The performance of the radiosonde in flight shall be monitored by ground station receiving equipment to determine that the radiosonde is performing in accordance with the requirements of this specification. Compliance with the

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specification shall be determined for the requirements listed below:

Frequency stability (403 MHz)	3.10.1.1.2
Frequency stability (1680 MHz)	3.10.1.2.2
Pulse width and rise time	3.10.2.2
Modulation frequencies	3.10.2.3
Frequency stability	3.10.2.4
Harmonic content	3.10.2.8
MDO mid-scale correction	3.10.2.9
Activation time	3.10.3.1
Commutator sequence	3.10.11.1
Flight termination	3.10.11.5
Baroswitch punched paper tape calibration	3.10.13
Commutator sequence	3.10.16.1
Commutator ON time	3.10.16.2

The contractor shall have access to the ground station during all flight tests.

4.5.5.3.2 Performance measurements during battery test. Battery voltage versus operating time shall be measured as specified in 4.7.3.1 and 4.7.3.2.

4.5.5.4 Accept/reject criteria.

4.5.5.4.1 First article criteria. The flight testing shall continue until seven relevant radiosonde failures occur (reject decision) or the 93 failure free flights are performed (accept decision). For purposes of this test, malfunction of the balloon/launch equipment or the ground station receiving/monitoring equipment is not a relevant failure. Also non-relevant are failures detected prior to launch and failures attributable to the crew. For example, breakage of elements during launch caused by radiosonde hitting the ground or object. All 45 sample batteries must meet the activation time and minimum time to end voltage requirements of 3.10.3.1 and 3.10.3.2 to demonstrate compliance with 3.12.2.

4.6 Weight and size. The Radiosonde Sets shall be weighed and measured with a calibrated scale to determine compliance with the requirement of 3.9.

4.7 Performance testing. The Radiosonde Sets verification testing shall be conducted in accordance with a contractor prepared, government approved test procedure required by the contract. The tests described by this procedure shall verify the correct functioning of each system and

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subsystem within its prescribed electrical and operational performance limits as follows (see 3.10).

4.7.1 Transmitters. The meteorological data frequency deviation (150 ± 30 KHz) and the NAVAID frequency deviation (200 ± 40 KHz), when used, shall be verified over a temperature range of -95°F (-70°C) to 125°F (51.7°C) and a voltage range of 15.0 to 23.0 volts. The measurements shall be made using spectrum analyzer or equal having a minimum resolution of 10 KHz at 403 MHz and 1680 MHz. Frequency of measurements and voltage and temperature profiles shall be in accordance with 4.7.11. (see 3.10.1)

The following tests shall be performed also to complete performance testing of the transmitters:

4.7.1.1 403 MHz output.

4.7.1.1.1 Frequency. The transmitter tuning range (400-406 MHz) and setting accuracy (+0.025 MHz) shall be verified at ambient temperature and nominal voltage (± 23.0 ± 0.5 volts) using an electronic frequency counter with an accuracy of at least 1.0 KHz. (see 3.10.1.1.1)

4.7.1.1.2 Frequency stability. The frequency stability (remain within a 1.0 MHz band) of a transmitter tuned to any frequency (fo) within the specified frequency range (400-406 MHz) shall be verified over a temperature range of -95°F (-70°C) to + 125°F (51.7°C) and a voltage range of 15 to 23.0 volts. The measurements shall be made using an electronic frequency counter with an accuracy of 1.0 KHz or better. Frequency of measurements and voltage and temperature profiles shall be in accordance with paragraph 4.7.11. (see 3.10.1.1.2)

4.7.1.1.3 Transmitter radiated power output. The radiated power output (200 MW Min.) of the transmitter tuned to any frequency within the specified frequency range (400-406 MHz) shall be verified over a temperature range of -95°F(-70.0°C) to 125°F (51.7°C) and a voltage range of 15.0 to 23.0 volts. The measurement shall be made using a power meter having a one percent accuracy or better. To facilitate making the test, effective radiated power shall be measured by connecting the power meter through a 50 ohm matching network directly to the transmitter antenna connector. Frequency of measurements and voltage and temperature profiles shall be in accordance with paragraph 4.7.11. (see 3.10.1.1.3)

4.7.1.1.4 Antenna. The match (1.5 to 1 max VSWR) between the transmitter output and the antenna over the specified frequency range (400-406 MHz) shall be verified at ambient temperature using a

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slotted-line VSWR bridge. (see 3.10.1.1.4)

4.7.1.1.5 Superegeneration. Nonsuperegeneration of the transmitter shall be verified over the temperature range of -95°F (-70°C) to ± 125°F (51.7°C and voltage range of 15.0 to 23.0 volts using a spectrum analyzer and viewing, as minimum, a sweep range of 200 to 600 MHz. Voltage and temperature profiles shall be in accordance with 4.7.11. (see 3.10 1.1.5)

4.7.1.2 1680 MHz output test.

4.7.1.2.1 Frequency. The transmitter tuning range (1600-1700 MHz) and setting accuracy (± 0.025 MHz) shall be verified at ambient temperature and nominal voltage (18.0 ± 0.5 volts) using an electronic frequency counter with an accuracy of at least 1.0 KHz. (see 3.10.1.2.1)

4.7.1.2.2 Frequency stability. The frequency stability (remain within a 2.0 MHz band) of a transmitter tuned to any frequency (fo) within the specified frequency range (1660-1700 MHz) shall be verified over a temperature range of -95°F (-70°C) to + 125°F (51.7°C) and a voltage range of 15.0 to 23.0 volts. The measurement shall be made using an electronic frequency counter with an accuracy of 1.0 KHz or better. The r-f coupling between the radiosonde and frequency counter shall be by direct connection through a 50 ohm directional coupler. Frequency of measurements and voltage and temperature profiles shall be in accordance with paragraph 4.7.11. (see 3.10.1.2.2.)

4.7.1.2.3 Transmitter radiated power output. The radiated power output (200 MW Min.) of the transmitter turned to any frequency within the specified frequency range (1660-1700 MHz) shall be verified over a temperature range of -95°F (70.0°C) to +125°F (51.7°C) and a voltage range of 15.0 to 23.0 volts. The measurement shall be made using a power meter having a one percent accuracy or better. To facilitate making the test, effective radiated power shall be measured by connecting the meter through a 50 ohm directional coupler directly to the transmitter antenna connector. Frequency of measurements and voltage and temperature profiles shall be in accordance with paragraph 4.7.11. (see 3.10.1.2.3)

4.7.1.2.4 Antenna. The match (1.5 to 1 max VSWR) between the transmitter output and the antenna over the specified frequency range (1600-1700 MHz) shall be verified at ambient temperature using a sloted-line VSWR bridge. (see 3.10.1.2.4)

4.7.1.2.5 Superegeneration. Nonsuperegeneration of the transmitter shall be verified over the temperature range of -95°F (-70°C) to

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+ 125°F (+51.7°C) and voltage range of 15.0 to 23.0 volts using a spectrum analyzer and viewing, as a minimum a sweep range of 1200 to 2200 MHz. Voltage and temperature profiles shall be in accordance with 4.7.11 (see 3.10.1.2.5).

4.7.2 Modulator.

4.7.2.1 Leakage resistance (modulator). The leakage resistance (open circuit frequency <1.0 Hz) of the modulator at high humidity shall be verified at nominal operating voltage (18.0 ± 0.5 volts). With no temperature element in the circuit the measurement shall be made by viewing the meteorologic data oscillator output on an oscilloscope and measuring the period of the leakage pulses. The measurement shall first be made at ambient temperature and humidity after operating the modulator for 10 minutes. Then the temperature shall be raised to 40°C ± 5°C and the humidity to 90 + 5, -0 percent. The open circuit frequency shall be determined after operating in this environment for 2 hours. (see 3.10.2.6).

4.7.2.2 Modulation frequencies. The meteorological data oscillator frequency range (12.5 to 450 Hz and at 475 Hz for reference) shall be verified at ambient temperature and nominal voltage (18.0 ± 0.5 volts) using an electronic frequency counter with a minimum accuracy of ± 0.1 Hz. Appropriate resistance values shall be inserted into the temperature sensor circuit. (see 3.10.2.3)

4.7.2.3 Frequency stability. The meteorological data oscillator frequency stability (± 20 Hz) at reference frequency (475 Hz) shall be verified over the temperature range of -95°F (-70.0°C) to 125°F (51.7°C) and voltage range of 18.0 to 23.0 volts. The measurement shall be made using an electronic frequency counter with a minimum accuracy of ± 0.1 Hz. Frequency of measurements and voltage and temperature profiles shall be in accordance with 4.7.11. (see 3.10.2.4)

4.7.2.4 Transfer characteristic. The transfer characteristic shall be verified over the temperature range of -95°F (-70.0°C) to 125°F (51.7°) and a voltage range of 18.0 to 23.0 volts. The measurement shall be made using an electronic frequency counter with a minimum accuracy of ± 0.1 Hz. The specified resistors shall have resistance value within ± 0.1 percent tolerance and shall be inserted one at a time into the temperature sensor circuit and the resulting frequencies measured. The mid-scale frequency shall also be measured. For each measured frequency, compute from Equation 1 and 2 the equivalent sensor resistance. Next, compute the temperatures, using Equation 3, for the specified resistance and the computed equivalent sensor resistance. The temperature difference is the transfer characteristic accuracy.

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Frequency of measurements and voltage and temperature profiles shall be in accordance with 4.7.11. (see 3.10.2.7)

4.7.2.5 Pulse width and rise-time. The pulse width and rise time shall be verified over a temperature range of -95°F (-70.0°C) to 125°F (51.7°C) and a voltage range of 15.0 to 23.0 volts. The pulse width measurement shall be measured using an electronic frequency/period counter with an accuracy of ± 0.1 Hz ± 5 microseconds. The rise time shall be measured using a calibrated oscilloscope. Frequency of measurements and voltage and temperature profiles shall be in accordance with paragraph 4.7.11. (see 3.10.2.2)

4.7.2.6 Sensor current. The meteorological data oscillator sensor current (<75 microamperes dc) shall be verified at ambient temperature and maximum voltage (23.0 volts), using and oscilloscope to measure the peak voltage across a fixed resistor connector in the temperature circuit. A 6800 ohm resistor (5 percent tolerance) shall be connected across the temperature mounting clips and the peak voltage across the resistor shall be measured with the oscilloscope. The average current shall be calculated. (see 3.10.2.5)

4.7.2.7 Mid-scale correction. The mid-scale correction resistor value shall be verified by measuring its resistance with a resistance bridge. As an alternate method, the meteorological data oscillator frequency can be measured with the baroswitch contact on an insulated segment. The frequency shall be one-half the reference frequency. (see 3.10.2.9)

4.7.2.8 Harmonic content. The effects of the harmonic content from the meteorological data oscillator output; including harmonics from the clock communicator, on the processed NAVAID data shall be verified. If a Meteorological Data System AN/TMQ-31 is available, measurements shall be made during the test flights conducted under group D inspection (4.5.5.1). The processed NAVAID data shall be reviewed for indications of erroneous data. Without the availability of an AN/TMQ-31, the measurement shall be made by viewing on an oscilloscope the demodulated NAVIAD signal for evidence of meteorological data oscillator harmonics in the various NAVAID pass bands. (see 3.10.2.8.)

4.7.3 Radiosonde battery.

4.7.3.1 Activation time. The time to activate the battery (10 min) shall be verified at ambient temperature. The measurement shall be made using a dummy load, which simulates the radiosonde load to within ± 1 percent with a supply voltage of 18.0 volts, and a digital voltmeter with an accuracy of ± 0.3 percent. The battery shall be activated in

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accordance with the instruction provided with it and then placed underload. The activation time shall be measured from when the battery is connected to the load to when the battery voltage reaches 15.0 volts (minimum radiosonde operating voltage). (see 3.10.3.1)

4.7.3.2 Minimum time to end voltage. The minimum time to battery end voltage (>120 min.) shall be verified. The measurement shall be made using an environmental chamber with a capability of simulating the atmospheric pressure-temperature profile at a rate of 1000 ± 50 feet/minute from surface to 105,000 feet. The voltage shall be measured with digital voltmeter with an accuracy of 0.3 percent. The battery shall be activated in accordance with the instructions provided with it and placed in the chamber under the load specified in 4.7.3.1. For the first 10 minutes, the chamber shall remain at ambient temperature and surface pressure. After 10 minutes, the chamber pressure and temperature shall be changed in accordance with the US Extension to the ICAO Standard Atmosphere at a rate of 1000 ± 50 feet/minute. Voltage measurements shall be recorded every 5 minutes for the 120 minute period. The time required for the battery to decrease to the minimum voltage of 15.0 volts shall be determined. (see 3.10.3.2)

4.7.4 Receiver. The receivers shall be tested as follows:

4.7.4.1 LORAN C/D.

4.7.4.1.1 Antenna. The match (1.5 to 1 Max VSWR) between the receiver input and the antenna shall be verified at ambient temperature. (see 3.10.4.1.1)

4.7.4.1.2 Amplifier gain/bandwidth. The amplifier gain/bandwidth (>40dB/10.0KHz at -3 dB) shall be verified over a temperature range of -95°F (-70.0°C) to + 160°F (71.1°C) and at voltage range of 15.0 to 23.0 volts. Measurements shall be made using a waveform generator, oscilloscope, variable attenuator, and frequency counter. Input to the receiver shall be through the antenna connector, the waveform generator output shall be sinusoidal, and the waveform generator frequency shall be measured with a frequency counter having an accuracy of ±1.0Hz. The amplifier input voltage frequency shall be adjusted to 100 KHz ± 0.5 KHz (fo) and a standard voltage level shall be selected. This level shall not cause receiver saturation when attenuated by 40 dB and fed to the amplifier input. Bandwidth shall be determined by adjusting the amplifier input voltage to produce an output equal to the standard voltage level at fo, and measuring the frequencies at which the output voltage becomes -3 dB and -20 dB of the standard level. Gain shall then be determined at the -3 dB frequencies by measuring the amount of attenuation required to obtain a standard voltage level at the amplifier

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output when applying this voltage level to the input through an attenutor. Frequency of measurements and voltage and temperature profiles shall be in accordance with 4.7.11. (see 3.10.4.1.2)

4.7.4.2 OMEGA/VLF

4.7.4.2.1 Antenna. The match (1.5 to 1 Max VSWR) between the receiver input and the antenna shall be verified at ambient temperature. (see 3.10.4.2.1)

4.7.4.2.2 Amplifier gain/bandwidth. The amplifier gain/bandwidth (>40dB/17.5 + 2.5 KHz at -3 dB) shall be verified over a temperature range of -95°F (-70.0°C) to + 125°F (51.7°C) and at voltage range of 15.0 to 23.0 volts. Measurements shall be made using a waveform generator, oscilloscope, variable attenuator, and frequency counter. Input to the receiver shall be through the antenna connector, the waveform generator output shall be sinusoidal, and the waveform generator frequency shall be measured with a frequency counter having an accuracy of ± 1.0 Hz. The amplifier input voltage frequency shall be adjusted to 17.0 ± 0.1 KHz (fo) and a standard voltage level shall be selected. This level shall not cause receiver saturation when attenuated by 40 dB and fed to the amplifier input. Bandwidth shall be determined by adjusting the amplifier input voltage to produce an output equal to the standard voltage level at fo, and measuring the frequencies at which the output voltage becomes -3 dB and -20 dB of standard level. Gain shall then be determined at the -3 dB frequencies by measuring the amount of attenuation required to obtain a standard voltage level at the amplifier output when applying this voltage level to the input through an attenuator. Frequency of measurements and voltage and temperature profiles shall be in accordance with 4.7.11. (see 3.10.4.2.2).

4.7.5 Temperature sensor tests. The temperature sensor shall be tested in accordance with MIL-T-49447 (CR) paragraph 4.7.

4.7.6.1 Baroswitch deflection. The baroswitch deflection shall be verified by placing the modulator in the pressure chamber and determining the increment between the make of successive contacts on decreasing pressure in the 1000 mb and 50 mb pressure regions (see 3.10.5.3).

4.7.6.2 Setting error. The setting error shall be verified by placing the modulator in the pressure chamber and observing the true pressure at the make of the 6th (R3) contact with decreasing pressure. The pressure difference between the calibration chart value and the true pressure, is the setting error. (see 3.10.5.4)

4.7.6.3 Calibration accuracy. The calibration accuracy shall be

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verified. With the baroswitch contact arm lowered on the commutator, the baroswitch assembly shall be placed in a test chamber and electrical connections shall be made to indicating instruments external to the chamber. The pressure chamber temperature shall be maintained at $27^{o}C \pm 7^{o}C$ and the pressure shall be increased to a pressure between 1060 and 1030 mb. The pressure shall then be reduced at as fast a rate as possible, except while making observations, when the rate of change of pressure not exceed 1 mb in 5 seconds. The true pressure in the chamber shall be observed at the make of the sixth contact and at least 15 other contacts equally spaced to a maximum pressure of 20 mb. The pressure shown on the calibration, after correction for setting error as described in 3.10.5.4, shall be compared to the observed pressure. (see 3.10.5.5)

4.7.6.4 Low temperature effect. The calibration accuracy at low temperature shall be verified. After the test of 4.7.6.3 and without adjustment or resetting, the baroswitch assembly shall be tested at $80^{o}C \pm 3^{o}C$ below that used for test 4.7.6.3. The assembly shall be placed in the test chamber and brought to temperature equilibrium, after which the pressure shall be reduced and pressure observation taken at the same contacts taken in 4.7.6.3. The true pressure for each observed contact at low temperature shall be compared to the true pressure for the same contact at room temperature taken under 4.7.6.3 (see 3.10.5.6).

4.7.6.5 Friction. The friction shall be verified by placing the modulator in the pressure chamber and recording true pressure for each R6 contact (No. 2, 16, 30, etc.) observed at the make of the contact during decreasing pressure and the break for increasing pressure. The difference in the two pressures is the friction. (see 3.10.5.7)

4.7.6.6 Detent operation. The detent operation shall be verified at ambient pressure as follows:

a. The detent adjustment screw shall be turned and the maximum number of commutator segments traversed by the contact point noted.

b. The detent adjustment screw shall be turned so that the contact point makes contact with the leading edge of the 6th commutator segment. The adjustment screw shall be turned clockwise until the contact point makes contact with the leading edge of the 7th commutator segment. During this adjustment, the number of detent clicks shall be counted. Using the calibration chart, the detent click value (mb/click) shall be determined.

c. With the contact arm point at the 7th commutator segment, the detent adjustment screw shall then be turned in "five click

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intervals" in a clockwise direction for fifteen clicks, and "five click intervals" in a counter-clockwise direction for fifteen clicks. From this contact arm position the number of clicks required, if any, to position the contact arm point back to the leading edge of the 7th commutator segment shall be determined. (see 3.10.6)

4.7.6.7 Contact pressure. The contact resistance (<10 Ohms) shall be verified at ambient pressure using a spring gage, with the baroswitch in an upright position and the contact arm point centered on a commutator segment. A continuity indicator shall be used to indicate when the contact arm point breaks contact with the commutator segment. (see 3.10.8)

4.7.6.8 Contact resistance. The contact pressure shall be verified using a digital ohmmeter having an accuracy of ± 0.3 percent. Contact resistance shall be measured in the 1000 mb pressure region with the contact arm point centered on a commutator segment and with normal operating contact bearing pressure. (see 3.10.9) This test shall be conducted on units which have passed the contact pressure test 4.7.6.7.

4.7.7 Commutator tests.

4.7.7.1 Commutator sequence. The commutator sequence and resistance values shall be verified at ambient temperature by measuring the respective frequency of an meteorological data oscillator for each resistive commutator segment connected to the input and computing the equivalent resistance by using Equations 1 and 2 of 3.10.2.7. An electronic frequency counter shall be used to measure the frequency and shall have an accuracy of ± 0.1 Hz. As an alternate method, the resistance can be measured directly with a digital ohmmeter having an accuracy of +0.3 percent. (see 3.10.11.1)

4.7.7.2 Adherence of conducting segments. Adherence of conducting commutator segments shall be verified at ambient temperature and pressure. A strip of pressure sensitive tape shall be applied over the printed portion of the commutator with thumb pressure and then pulled slowly (as differentiated from a quick pull). Upon removal of the tape the commutator shall be examined for evidence of loosened or removed portions of the printed circuit. (see 3.10.11.2)

4.7.7.3 Leakage resistance (commutator). The leakage resistance of the commutator shall be verified. The baroswitch assembly shall be placed for 2 hours in a chamber with a relative humidity of 90 + 5, -0 percent and a temperature of 40°C ± 5°C. Immediately there after, the resistance shall be measured separately between metallic

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ground (baroswitch assembly main-frame) and the common even-numbered contact eyelet and the main-frame and the common odd-numbered contact eyelet and between the two eyelets with a megohm bridge having and accuracy of ± 4 percent. (see 3.10.11.3)

4.7.7.4 Segment width and thickness. The commutator segment width and thickness shall be verified by means of a micrometer microscope or equal. (see 3.10.11.4)

4.7.7.5 Flight termination. The minimum and maximum flight termination altitudes shall be verified by placing the shorting pin first in hole A and then K on the commutator and determining the contact number that the contact point rests on when the contact arm is moved against the shorting pin. Using the calibration chart for determining the pressure, the altitude shall be determined by using the US Extension to the ICAO Standard Atmosphere. (see 3.10.11.5)

4.7.7.6 Baroswitch calibration chart. The baroswitch calibration chart shall be examined to determine compliance with 3.10.12.

4.7.7.7 Baroswitch punched paper tape calibration. The baroswitch punched paper tape calibration shall be examined to determine compliance with 3.10.13.

4.7.8 Electronic commutator tests.

4.7.8.1 Commutator sequence. The commutator sequence (P,T,P,H,P,T,P,R) shall be verified. The measurement shall be made at ambient temperature by recording the commutated meteorological data oscillator output on a strip chart recorder, or equal and checking the format. (see 3.10.16.1)

4.7.8.2 Commutator ON time. The commutator ON time (1 ± 0.1 seconds) shall be verified over a temperature range of - 95°F (-70°F) to + 125°F (51.7°C) and a voltage range of 15.0 to 23.0 volts. The measurement shall be made by recording the commutated meterorological data oscillator output on a strip chart recorder, or equal, having a feed speed of at least 50mm per second and measure the time from the recording. Frequency of measurements and voltage and temperature profiles shall be in accordance with 4.7.11. (see 3.10.16.2)

4.7.8.3 Commutator switch resistance variation. The maximum variation in ON resistance (40 ohms) shall be verified at ambient temperature and nominal voltage (18.0 ± 0.5 volts). The measurement shall be made using an electronic frequency counter with a minimum accuracy of ± 0.5 Hz. Electrical shorts shall be made across the

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temperature and humidity support clips. Temperature, humidity reference, and mid-scale frequencies shall be measured and recorded. Equations 1 and 2 from paragraph 3.10.2.7 shall be utilized to compute the resistance variations. (see 3.10.16.3)

4.7.9 Instrument case tests.

4.7.9.1 Reflectivity. The reflectivity to solar radiation in the visible light region shall be verified by comparing the amount of light reflected from a freshly scraped magnesium carbonate block with the amount reflected from the modulator case with a reflectometer. (see 3.10.14.1)

4.7.9.2 Humidity sensor housing. The reflectivity to solar radiation in the 0.30 to 4.0 micron energy band shall be verified by measuring the absorbance factor. (see 3.10.14.3)

4.7.9.3 Humidity element mounting. The percentage of airflow to which the humidity element is exposed shall be verified. The airflow across the humidity element in the humidity duct and that above the radiosonde shall be measured in a wind tunnel. The airflow measurements above the radiosonde shall be made at a sufficient distance to minimize effects of turbulance. (see 3.10.14.5)

4.7.9.4 Strength of case, ring and harness. Case harness and ring strength shall be verified using a spring scale by applying a constant 25 pound pull to the radiosonde. (see 3.10.14.4)

4.7.10 Temperature element clips.

4.7.10.1 Holding capability of clips. The holding capability of the clips shall be verified by inserting a temperature element in the temperature element clips and connecting a suitable spring scale to the temperature element lead close to 1 clip. A force shall be exerted in a direction which would most easily disengage the element lead from the clip. (see 3.10.7.1)

4.7.10.2 Leakage resistance (support arms). The leakage resistance of the support arms shall be verified. The modulator shall be placed for 2 hours in a chamber with a relative humidity of 90 + 5, -0 percent and a temperature of 40°C ± 5°C. Immediately thereafter, the leakage resistance shall be measured between the clip and the low end of the support arm. This measurement shall be made for both support arms using a megohm bridge having an accuracy of ± 4 percent. (see 3.10.14.7.2)

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4.7.11 Temperature test for individual performance requirement other than at ambient temperature.

a. Operate the radiosonde from a power supply at nominal battery voltage and ambient temperature for 10 minutes.

b. Raise the temperature rapidly to + 125°F (+5°F, -0°F), 51.7°C (+2.8°C, -0°C). Following a 10 minutes period, reduce the temperature to a minimum value (as determined in paragraph 4.7.11.1) in steps of 10°C per 8 minutes or fraction therefor the last step, and allow another 10 minutes for stabilization at minimum temperature.

c. Measurements shall be taken every 8 minutes, coinciding with the temperature steps in b above for supply voltages of 15, 18 and 23 volts to determine compliance with specified requirements.

4.7.11.1 Method for determining minimum temperature. The minimum temperature shall be determined by operating the radiosonde with a battery in a temperature-pressure chamber and instrumenting the radiosonde to measure the internal free-air transmitter temperature. The method shall be as follows:

a. After the battery has reached minimum operating voltage, operate the radiosonde for 10 minutes at 15°C and ambient pressure.

b. Change the temperature and pressure conditions within the chamber in accordance with the temperature-pressure profile of table IV. The rate of change of pressure shall be equivalent to an altitude change of 1000 ± 50 ft (305 ± 15 meters) per minute.

c. Measure and record the temperature within the radiosonde transmitter every minute and note the coldest temperature measured.

d. Remove the battery from the radiosonde and allow radiosonde to stabilize at room temperature for at least 5 hours.

e. Connect a power supply, operating at the minimum battery voltage, to the radiosonde.

f. Operate the radiosonde at ambient pressure and in accordance with 4.7.11 a and b, except at minimum battery voltage, to

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determine the minimum chamber temperature that will produce an internal radiosonde transmitter temperature equal to that determined in 4.7.11. 1c, above.

Add table IV as follows:

TABLE VI Temperature - pressure profile.

ALTITUDE	PRESSURE	TEMPERATURE
KM	mb	°C
0	1013	15.0
1	931	11.3
2	829	4.5
3	736	0.9
4	653	-4.7
5	559	-11.0
6	492	-18.1
7	432	-26.1
8	378	-33.7
9	329	-41.1
10	286	-49.0
11	247	-55.8
12	213	-61.4
13	182	-67.0
14	156	-72.0
15	132	-77.8
16	112	-83.7
17	94	-87.0
18	79	-86.1
19	67	-81.4
20	56	-75.4
21	48	-71.0
22	41	-67.2
23	35	-65.0
24	30	-62.7
25	26	-61.0
26	22	-59.7
27	19	-59.1
28	16	-58.5
29	14	-57.9
30	12	-57.3

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4.7.12 Temperature ambient. Wherever ambient temperature is specified, the tests shall be conducted at 27°C ± 7°C.

4.8 Service conditions. The service condition tests shall be performed as detailed below. Where operation of the test item is required during any service condition test, the test item shall be required to pass the operational tests shown in table VII.

TABLE VII
Operational Tests for Group C Inspection

Operational Test 1/	Service Condition Test			
	Temperature Altitude 4.8.1	Humidity 4.8.2	Rain 4.8.3	Orientation 4.8.7
Transmitter (403 MHz) power output (4.7.1.1.3) 2/	X	X	X	X
Transmitter (1600 MHz) power output (4.7.1.2.3) 2/	X	X	X	X
Modulation frequency (4.7.2.2)	X	X	X	X
Amplifier gain/band width (Loran (4.7.4.1.2) 2/	X	X	X	X
Amplifier gain/band width OMEGA/VLF (4.7.4.2.2) 2/	X	X	X	X
Commutator Sequence (4.7.7.1) 2/	X			

1/ Operational tests are not required during the vibration test (4.8.4) and bench handling Test (4.8.5).

2/ The temperature condition shall be that as specified for the specified service condition test.

4.8.1 Temperature-altitude. The Radiosonde Sets shall be tested as follows: (see 3.11.1)

Step 1: Place the test item(s) in the chamber instrument as required and adjust the chamber conditions to standard

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conditions as specified in accordance with MIL-STD-810, Method 504.1 Procedure 1.

Step 2: Operate the test item(s) to determine if the performance is within specified limits.

Step 3 Change the chamber temperature to + 125°F (+51.7°C) and maintain for 4 hours following temperature stabilization.

Step 4: Repeat the test measurements of Step 2.

Step 5: Change the temperature and pressure conditions within the chamber in accordance with the US Extension to the ICAO Standard Atmosphere. The rate of change altitude shall be equivalent to 1,000 feet per minute. Operate radiosonde and record measurements every 5 minutes during the change from sea level conditions to those at 105,000 feet (32,000m).

Step 6: Change chamber conditions to standard, inspect for damage, and repeat Step 2.

4.8.2 Humidity test. The Radiosonde Sets (less temperature and humidity elements) shall be tested in accordance with Method 507.1, Procedure II of MIL-STD-810 except that step 6 shall consist of 2 cycles. (see 3.11.2)

4.8.3 Rain test. The Radiosonde Sets shall be tested in accordance with Method 506.1, Procedure I of MIL-STD-810. The Radiosonde Sets shall be reoriented every 30 minutes so that each side of the Radiosonde Sets will be equally exposed to simulated winddriven rain. Total test time shall be 2 hours. The Radiosonde Sets shall be operated during the last 10 minutes of each 30-minute period to determine if performance is within specified limits.

4.8.4 Vibration tests. The Radiosonde Sets shall be tested as follows:

a. Part 1. The Radiosonde Sets in its transit case if any, shall be subjected to the test of MIL-STD-810, Method 514.2, Procedure XI, Part 2. Prior to and following this test the Radiosonde Sets shall be operated in accordance with MIL-STD-810, general requirements, paragraph 3.2. The Radiosonde Sets shall not have any degradation in

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performance. The Radiosonde Sets including any transit case shall not suffer any mechanical damage.

b. Part II. The Radiosonde Sets in its transit case if any, shall be subjected to the test of MIL-STD-810, Method 514.2, Procedure X. The cycling and sweep time shall be in accordance with table 5142-VII. The vibration level shall be in accordance with figure 514.2-7 curve. Prior to and following this test, the Radiosonde Sets shall be operated in accordance with MIL-STD-810, general requirements paragraph 3.2. The equipment shall not have any degradation in performance. The Radiosonde Sets including any transit case shall not suffer any mechanical damage.

4.8.5 Bench handling test. The Radiosonde Sets shall be tested in accordance with Method 516.2, Procedure V of MIL-STD-810.

4.8.6 Fungus test. The Radiosonde Sets, units packed in as ready for delivery condition shall be subjected to the test of Method 508.1, Procedure I of MIL-STD-810 with the exception that the chamber conditions of paragraph 3.1.6(a) be as follows: Maintain the test chamber at 28°C (84°F) and 95 percent R.H. (minimum) during the life of the test. Keep the test chamber door closed during the incubation period except during inspection or for addition of other test items. The radiosonde or samples shall not be specially cleaned for the fungus test, except for the cleaning it received during or after production. After 28 days, based on visual examination, the radiosonde or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10 percent or less of the total area and no more than 6 unrelated minute colonies with noncritical development in noncritical areas. Critical areas include terminal spacing, printed circuit boards, etc. The radiosonde shall fail if it shows more than the growth specified above.

4.8.7 Orientation. The Radiosonde Sets shall be tested for compliance with the requirements of 3.11.7.

4.9 Visual and mechanical. The Radiosonde Sets shall be examined for the defects IAW Requirements 5 and 9 of MIL-STD-454 and conformance to the requirements of 3.8 and 3.10.15.2. An inspection shall be performed to verify compliance with those portions of 3.7 (Systems Safety Engineering) which can be determined visually.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the

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desired level(s) of protection shall be as specified by the acquisition activity.

6. NOTES. (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The Radiosonde Sets ML-659(V)1, ML-659(V)2 and ML-659(V)3 are the balloon-borne meteorological instruments for measuring and telemetering temperature, relative humidity and pressure data to the ground station for the RDF subsystem along with received NAVAID data for the NAVAID subsystem. The Radiosonde Sets shall consist of a temperature element, a humidity element, a pressure sensor, radiosonde housing, a battery, a transmitter (403 MHz and 1680 MHz; 403 MHz for NAVAID subsystem and 1680 MHz for RDF subsystem), and a NAVAID receiver when the NAVAID subsystem is utilized.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Marking and shipping of samples.

d. Place of final inspection.

e. Number of first articles to be submitted for approval (see 3.1).

f. Nameplate data.

6.3 Subject items (key word) list.

Transmitter (400-406 MHz)
Transmitter (1660-110 MHz)
LORAN/receiver
VLF/OMEGA receiver
Meteorological sensors
Temperature sensor
Humidity sensor
Pressure sensor or baroswitch assembly.

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This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of ALQs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

Custodian
Army - CR

Procuring activity:
Army - CR
Project 6660-A093

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-R-49315(CR)	Radiosonde Set ML-659(V) Family of

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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